

The Problem Statement

PS Number: SIH26189 — AI-Powered Criminal Network Analysis System

For: Ministry of Home Affairs (NCRB — National Crime Records Bureau)

Why This Problem Exists

Police and intelligence agencies collect huge amounts of information every day — FIR reports, phone call records, bank transaction records, surveillance notes, old criminal history, social media intel.

The problem: all this data is scattered across different systems and mostly unstructured (just paragraphs of text, not neat spreadsheets). When an investigator wants to figure out "who is this suspect actually connected to," they currently have to read through everything by hand and manually piece the connections together — slow, and it's easy to miss a hidden link buried in thousands of pages, especially when a criminal network is large and spread out.

What We're Building

A tool that reads all this scattered information automatically and draws the connections itself — like a web showing who's connected to whom — and points out the most important people in that web, instead of a human piecing it together manually.

How It Works, Step by Step

Step 1 — Read the reports and pull out names

We feed in text — an FIR report, a call log. The system reads it and automatically picks out the important pieces: person names, places, phone numbers, vehicle numbers, company names.

Example: *"Rahul met the owner of Sunil Traders in Sector 12, then called a number."*

The system pulls out: **Rahul** (person), **Sunil Traders** (organization), **Sector 12** (place), a **phone number**.

Step 2 — Connect the dots

Once we know the names, places, and numbers, we connect the ones that show up together. If Rahul and Sunil Traders appear in the same report, we draw a line between them. Do this across hundreds of reports, and a web of connections

naturally forms — dots (people, places, companies) joined by lines (their relationships).

Step 3 — Find who matters most

Not everyone in the web is equally important. Math formulas (already built into ready-made tools, nobody has to invent them) analyze the web and answer:

- "Who is connected to the most people?" — flags a possible key player.
- "Who connects two separate groups together?" — flags a possible middleman linking two gangs.

Step 4 — Store it

All these connections get saved in a database so they can be searched and looked up later.

Step 5 — Build the plumbing

Something has to connect all the pieces together — taking reports in, sending them through extraction, storing results, and serving them out to the screen.

Step 6 — Show it on screen

An investigator sees the web of connections drawn visually — clickable, searchable, filterable. Click a person, see who they're connected to. This is what actually gets shown live in the demo.

Why This Matters, In One Line

Instead of an investigator spending days manually connecting dots on paper, this tool does it in seconds and points straight at the people worth investigating first.

One Thing That Makes the Demo Actually Work

None of this looks impressive if the graph on screen is empty or boring. We're building realistic-looking fake reports with names and numbers that deliberately overlap, so when the system runs live, an actual interesting web shows up — not just three random dots.

The One-Sentence Pitch (everyone should be able to say this identically)

"We built a system that reads crime reports and automatically finds hidden connections between people, flagging the most important individuals in a criminal network — something investigators currently have to piece together by hand."

